

Differentiation test · Algebra

Grade 11

40 minutes

100 marks

17 questions + bonus

No calculator

Lessons 7–12 and 17–18, with limits at infinity

- Answer on separate paper, and show your working — method carries marks.
- Simplify every derivative. Leave no negative or fractional index in a final answer.

NAME

CLASS

DATE

MARK / 100

A

Section A · Four marks each

EASY

40 marks

Differentiate and simplify. One line of working is enough.

- 1

$y = 4x^3 - 5x + 7$ [4]
- 2

$y = \frac{3}{x^2}$ [4]
- 3

$y = 5\sqrt{x}$ [4]
- 4

$y = (2x + 1)(x - 4)$ [4]
- 5

$y = \frac{x^3 - 2x}{x}$ [4]
- 6

$y = (3x + 1)^5$ [4]
- 7

$y = (x^2 - 4)^3$ [4]
- 8

$y = \frac{1}{2x - 5}$ [4]
- 9

$y = \sqrt{4x + 1}$ [4]
- 10

Evaluate: $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x}{x^2 - 5}$ [4]

B

Section B · Eight marks each

MEDIUM

40 marks

Method marks are available even when the final answer is wrong.

- 11

Differentiate $y = \frac{2x - 1}{x^2 + 3}$ and find the gradient of the curve at $x = 1$. [8]
- 12

$y = x^3 - 3x^2 + 4$. Find the values of x at which the gradient of the curve is 9. [8]
- 13

$y = \sqrt{x^2 + 9}$. Find $\frac{dy}{dx}$ and its value at $x = 4$. [8]

14 Differentiate $y = \frac{5}{(1-2x)^3}$. [8]

15 Evaluate, showing the division in each case:

(a) $\lim_{x \rightarrow \infty} \frac{2x^2 - 3x + 1}{5x^2 + 4}$ [3]

(b) $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^3 - 2}$ [2]

(c) Explain why $\lim_{x \rightarrow \infty} \frac{x^3 + 2x}{4x^2 - 1}$ does not exist. [3]

C Section C • Ten marks each

Full method required.

HARD

20 marks

16 $y = x^2\sqrt{2x-1}$.

(a) Show that $\frac{dy}{dx} = \frac{x(5x-2)}{\sqrt{2x-1}}$. [6]

(b) Hence find the gradient of the curve at $x = 5$. [4]

17 The curve $y = \sqrt{2x+7}$.

(a) Find $\frac{dy}{dx}$. [3]

(b) Find the equation of the tangent at the point where $x = 1$. [4]

(c) Find the equation of the normal at that point, and the coordinates of the point where it crosses the x -axis. [3]

Bonus



Attempt this only when the rest is finished. It can make up marks lost elsewhere, but your total is still recorded out of 100.

BONUS

+10 marks

18 $y = \frac{x+1}{\sqrt{x}}$ for $x > 0$.

(a) Show that $\frac{dy}{dx} = \frac{x-1}{2x\sqrt{x}}$. [7]

(b) Hence find the point at which the tangent to the curve is horizontal. [3]